Workshop 1 Recap

Last time we defined Empathy as:

*"**A social and emotional skill** that helps us feel and **understand the emotions, circumstances, intentions, thoughts, and needs of others"** (McLaren, 2013)*

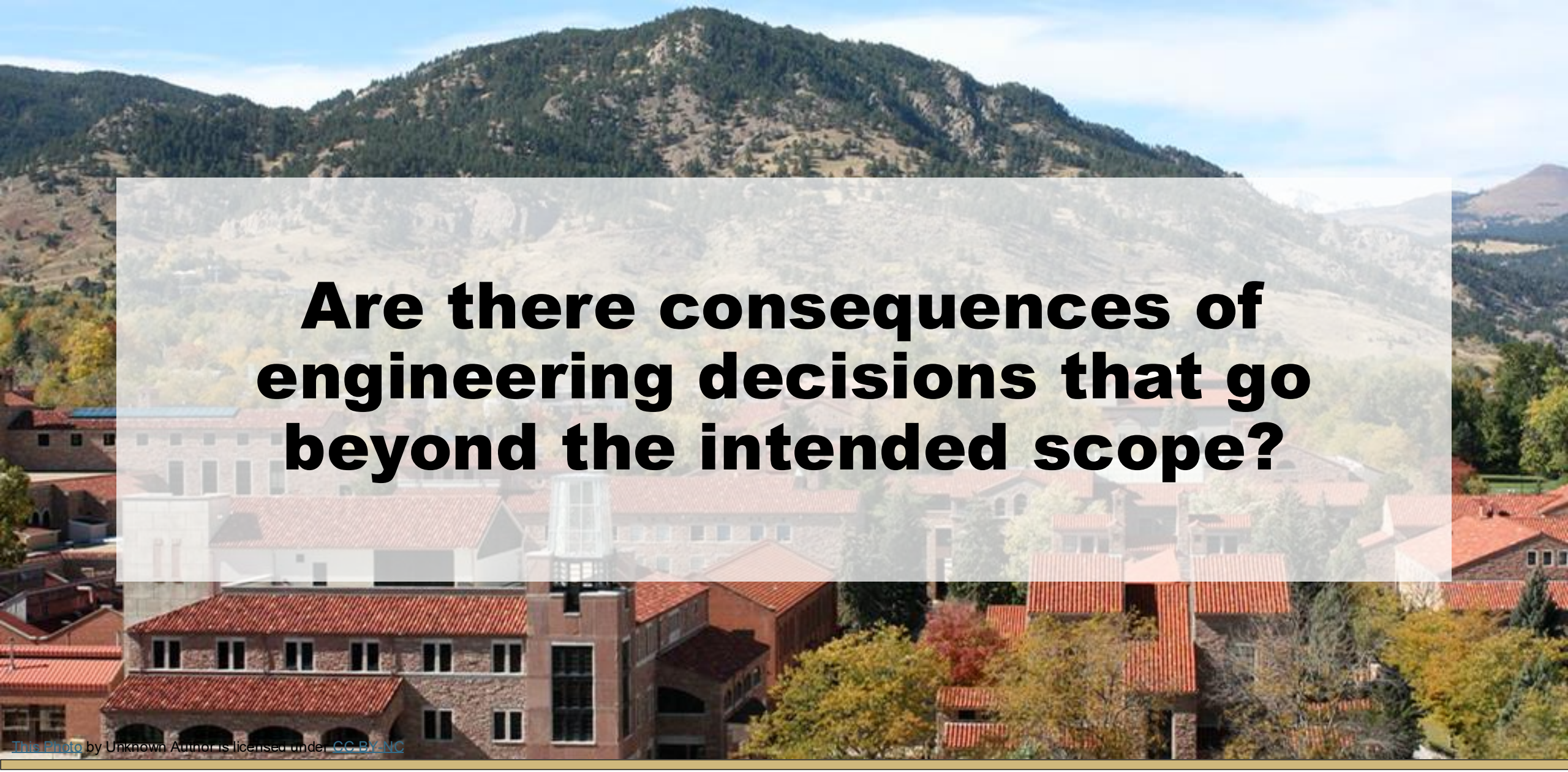


Last workshop:

- Used perspective taking (cognitive empathy)
- Identified stakeholders and their viewpoints

Today we'll build on that by asking:

- What consequences do engineering decisions create for different stakeholders?



Are there consequences of engineering decisions that go beyond the intended scope?

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Goals for the workshop today

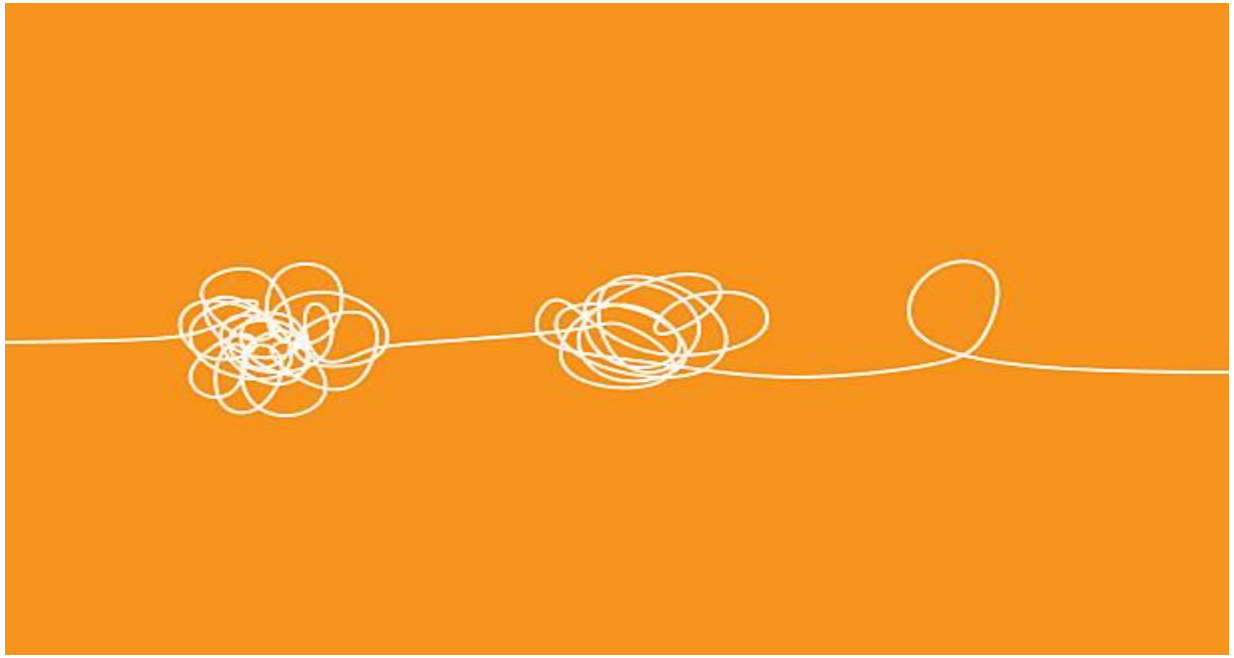
Use empathy and perspective taking to identify stakeholders.

Recognize how engineering decisions create different consequences for stakeholders.

Identify tradeoffs and consequences that may be overlooked during design.



Engineering Abstraction



Abstraction is the process of simplifying a complex problem so engineers can analyze and solve it.

While abstraction is necessary, it can make stakeholder perspectives and consequences easier to overlook.

(Schimpf, 2024)

Engineering Decisions Have Ripple Effects

A single technical decision can influence organizations, communities, people, and distinct groups in different ways.

Empathy helps engineers recognize these broader stakeholder consequences. (Walther, 2017)



Roadmap



Consequence Ladder

Debrief

Real-World Example

Reflection

Consequence Ladder

Consequence Ladder

Engineering decisions create consequences that extend far beyond the technical system.

You'll use the Consequence Ladder to trace how one decision affects different stakeholders across multiple levels of impact.



What is a Consequence?

A consequence is any effect or outcome that happens because of a decision.

Consequences can be:

- Positive or negative
- Immediate or long-term
- Intended or unintended
- Experienced differently by stakeholders



The Consequence Ladder

Engineering Scenario:

Your civil engineering team is helping a growing city reduce traffic congestion on a busy roadway.

To improve traffic flow, company has decided to r**emove the sidewalks to add an additional vehicle lane.**

As a team, determine the consequences of this decision.

02:00



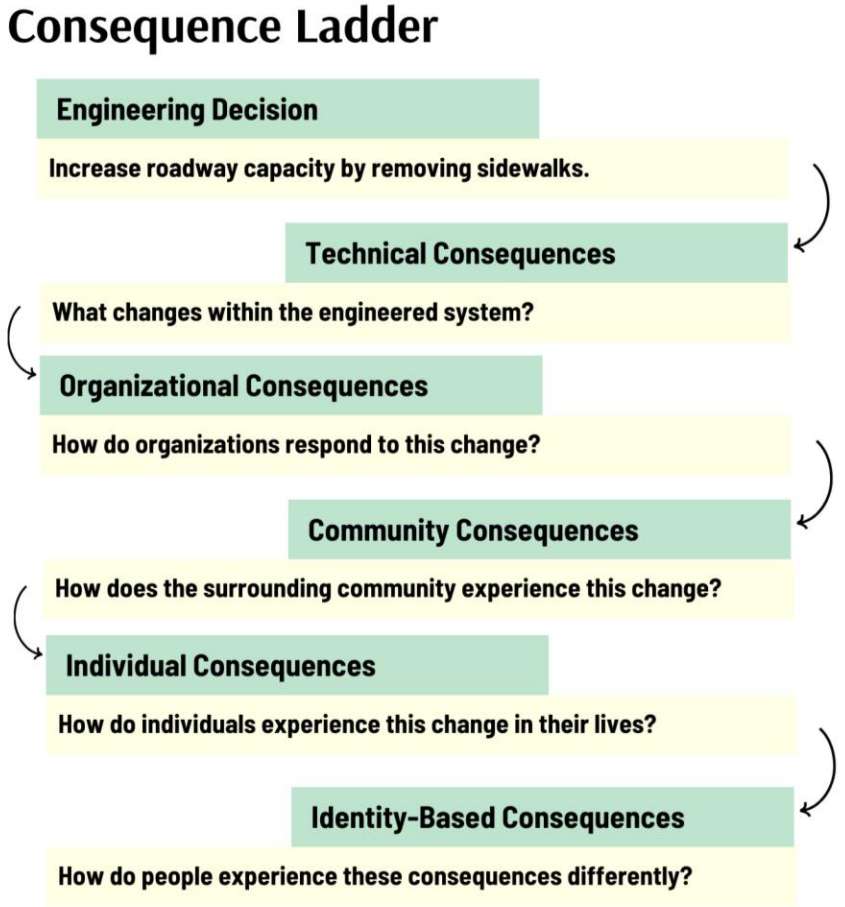
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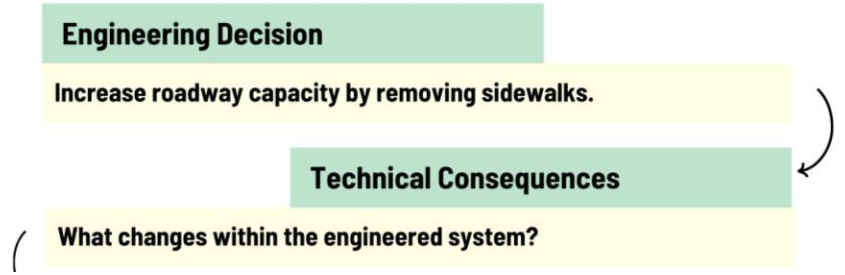
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Technical Consequences

What changes within the engineered system?



Think about changes to:

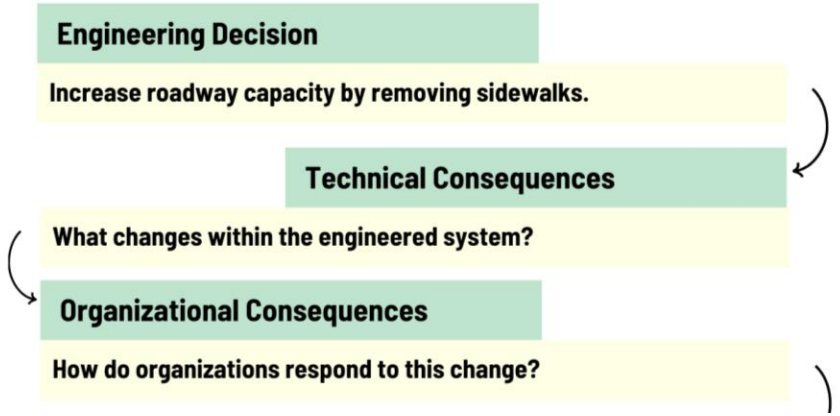
- Performance
- Capacity
- Safety
- Cost
- Reliability
- Environmental impacts

Example: Increased roadway capacity may reduce vehicle congestion.

02:00

Organizational Consequences

How do organizations respond to this change?



Consider organizations such as:

- City Government
- Maintenance departments
- Emergency responders
- Schools
- Businesses
- Company making the decision

Example: The city will spend less maintaining sidewalks.

02:00

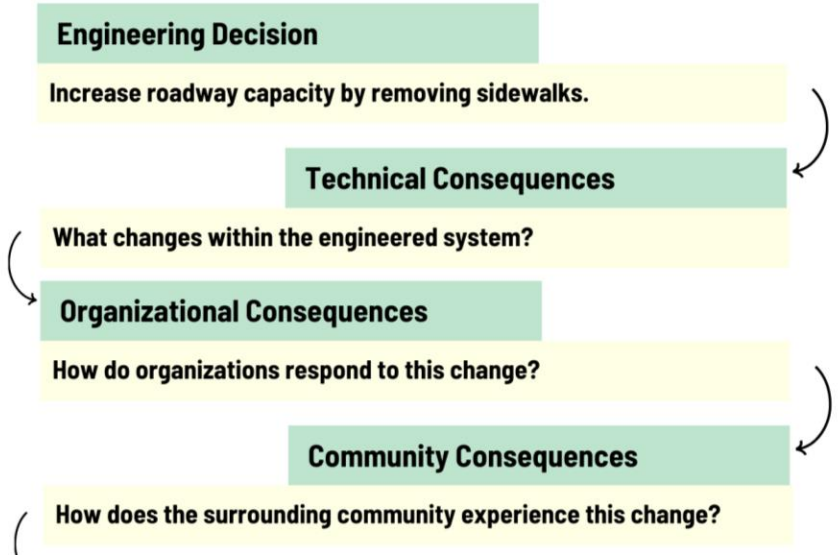
Community Consequences

How does the surrounding community experience this change?

Think about:

- Neighborhoods
- Local businesses
- Public spaces
- Community Life

Example: Local businesses see fewer pedestrians.



02:00

Individual Consequences

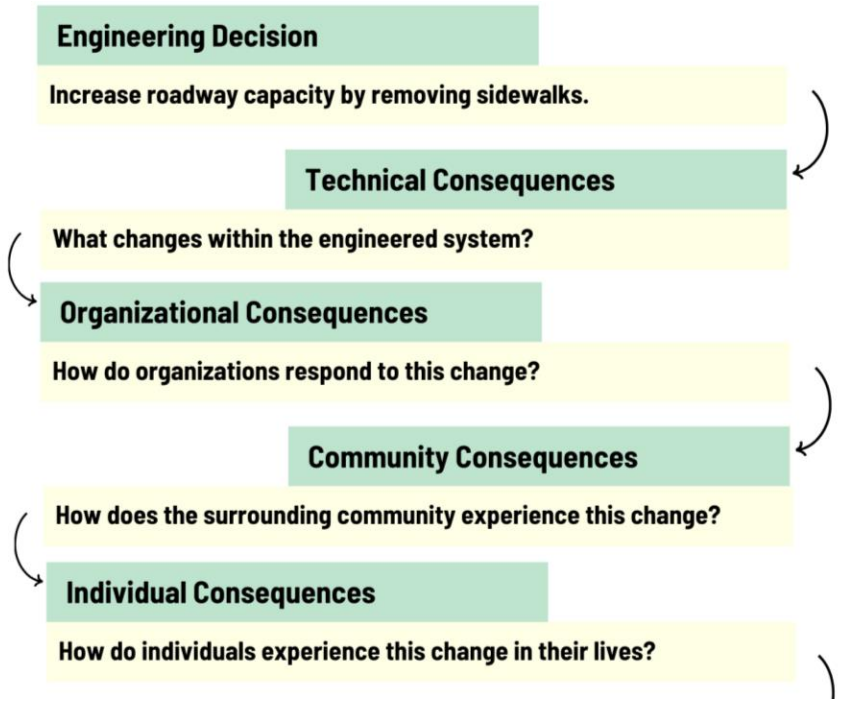
3:00

How do individuals experience this change in their lives?

Think about:

- Daily routines
- Safety
- Mobility
- Health
- Stress

Example: A nearby resident who walks to work now has a less safe route.



Identity-Based Consequences

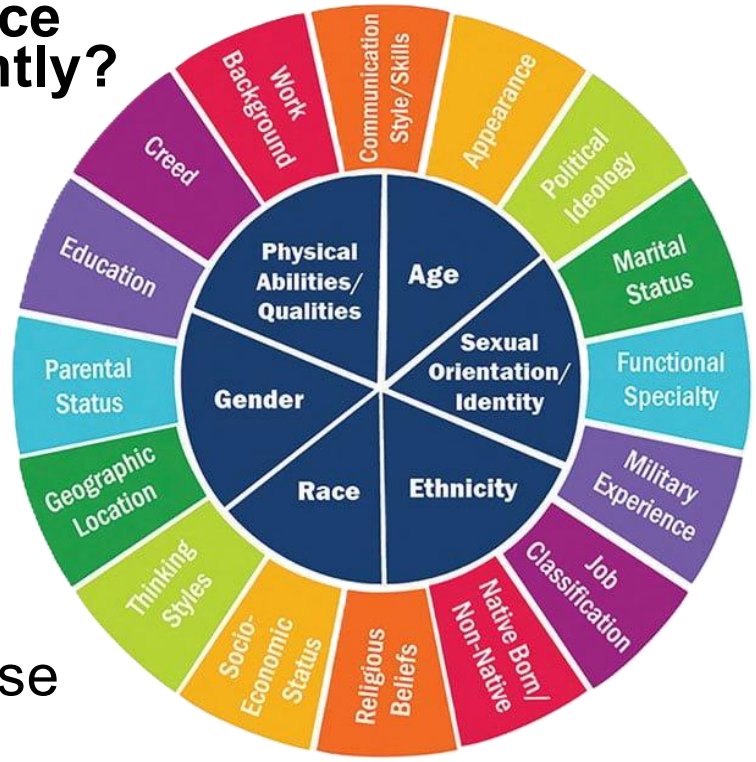
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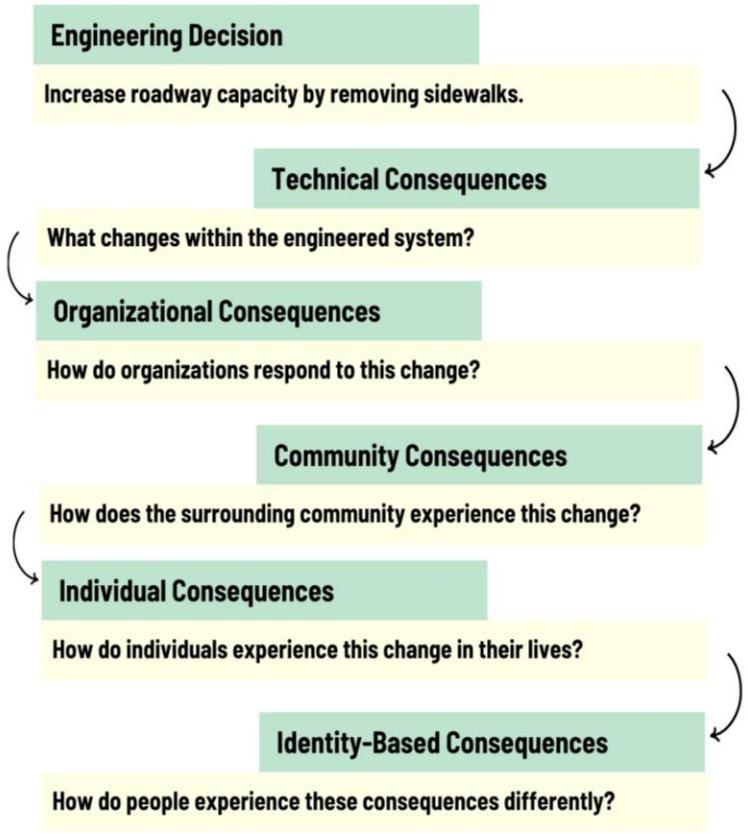
Do different people experience these consequences differently?

Think about:

- Age
- Disability
- Income
- Race

Example: Wheelchair users lose safe access.





Debrief

1. Which level was easiest to identify? Which was hardest? Why?
2. Which consequences do engineers usually consider? Which are often overlooked?
3. How did empathy or perspective taking help you identify additional stakeholder consequences?
4. If engineers stop at the technical level, what kinds of consequences become invisible?

Application In The Real World

What are Flock Safety Cameras?

- License Plate Recognition (LPR) cameras that automatically read and record vehicle license plates
- Captures: license plate number, vehicle description/color, location, date, and time
- Data is searchable and cross-referenced with law enforcement databases
- Helps law enforcement: solve crimes, find missing persons, respond to active alerts
- Reported to have helped solve thousands of crimes nationally



Your Task

You are engineers for Flock Safety. Boulder City Council has asked your team to recommend where to install License Plate Recognition cameras.

Your task: Use the Consequence Ladder to analyze two location options and advise the council on which location makes the most sense.

Stakeholders	Option A	Option B
Technical		
Organizational		
Community		
Individual		
Identity-Based		

Decision:

Why:

Area A: CU Boulder Campus and Surrounding Neighborhoods - Student dorms, parking, walking paths

WHY POLICE WANT CAMERAS HERE

- Deter assault and harassment on walking paths at night
- Provide evidence for crimes against students

GROUPS OF PEOPLE IN THIS AREA

- International students
- Unhoused individuals
- Students living off-campus
- Graduate and undergraduate students

AREA B: BOULDER CREEK & PUBLIC LIBRARY AREA - Outdoor creek area and public library

WHY POLICE WANT CAMERAS HERE

- Deter assault and harassment in the area
- Monitor for trespassing or loitering violations

GROUPS OF PEOPLE IN THIS AREA

- Students and families
- Swimmers and fishermen
- Elderly residents
- Tourists and visitors



Activity Debrief

Solution share out & reflection

Reflection

- Which location did you choose? What were the deciding factors?
- Which level of the ladder influenced your final decision the most? The least?
- Which levels of the ladder were hardest to fill out? Why? Was there anything that would have made it easier to fill out?
- Do engineers have responsibility for consequences beyond the technical system?
- How might today's workshop influence how you approach your next engineering project?

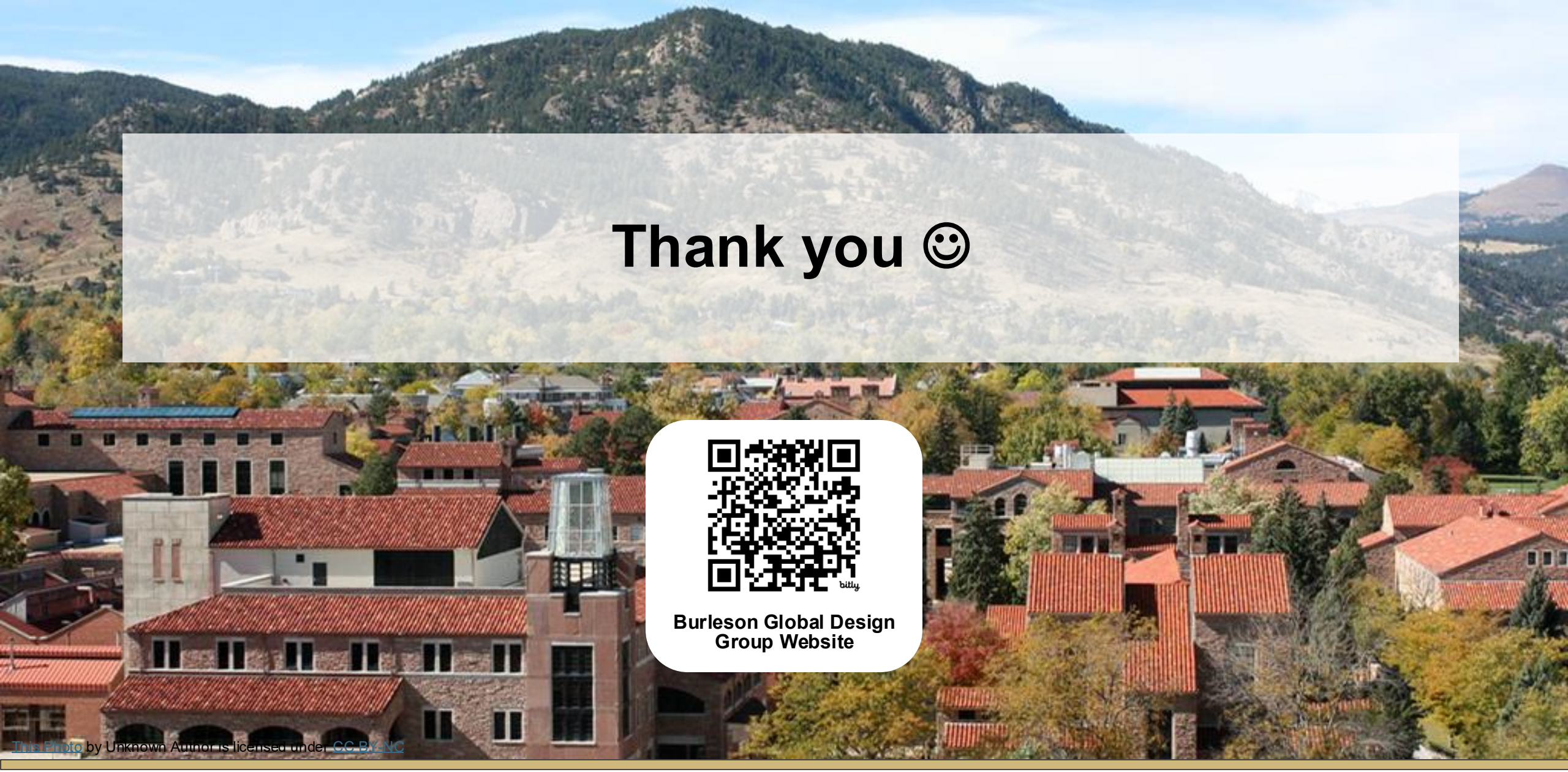
“Guidelines” (rename this)

- Do engineers have responsibilities? -> yes and no, sometimes its out of scope, but important to bring up concerns of the technical system
- Important to think of perspectives and how design may impact different people

Closeout

- **Engineering decisions have human consequences.** Technical choices create ripple effects that extend beyond performance and costs to organizations, communities, and individuals.
- **Empathy is a design tool, not just a personal value.** Considering multiple perspectives helps engineers identify hidden impacts and make more informed tradeoffs.
- **Ask who benefits, who bears the cost, and whose voice is missing.**

Thank you ☺



Burleson Global Design Group Website

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Feedback



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